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Parameter identification of viscoelastic materials using different deformation velocities

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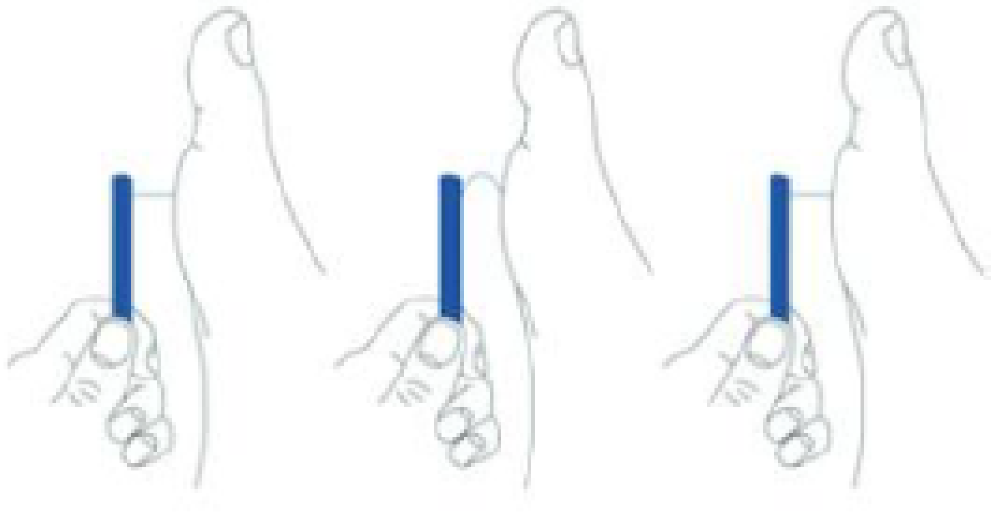
Outline:

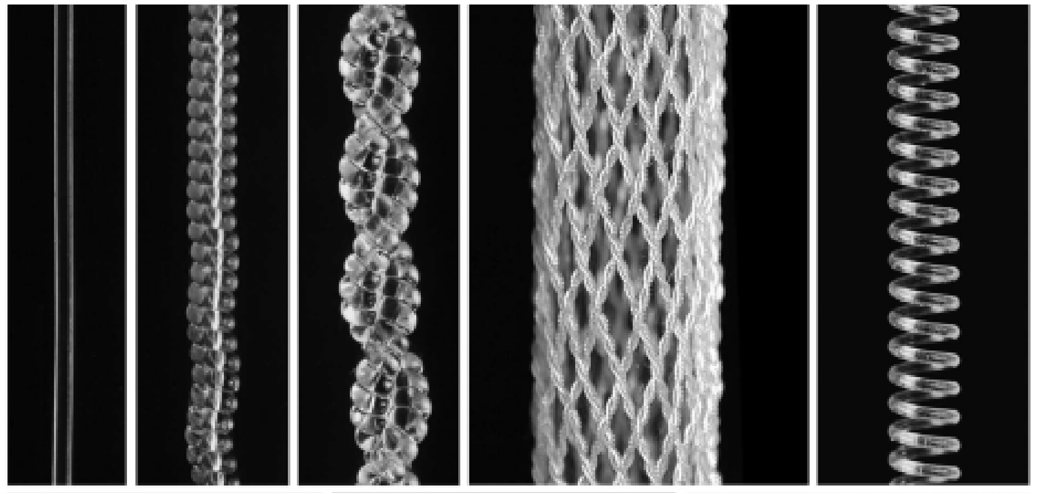
- Introduction
- Stress-relaxation experiment
- Fractional Kelvin- Zener model
- Results and conclusions

1.Introduction

- Monofilament made of fishing nylon has attracted special attention in medicine due to its mechanical properties.
- It was demonstrated that inexpensive high-strength polymer fibers used for fishing line and sewing thread can be easily transformed by twist insertion to provide fast, scalable, nonhysteretic, long-life tensile, torsional muscles.



Semmes Weinstein monofilament test



Artificial monofilament muscle

Monofilament fishing line

This paper investigates the influence of deformation rate on the values of parameters of the fractional Kelvin-Zener model.

2.Experiment and parameter identification

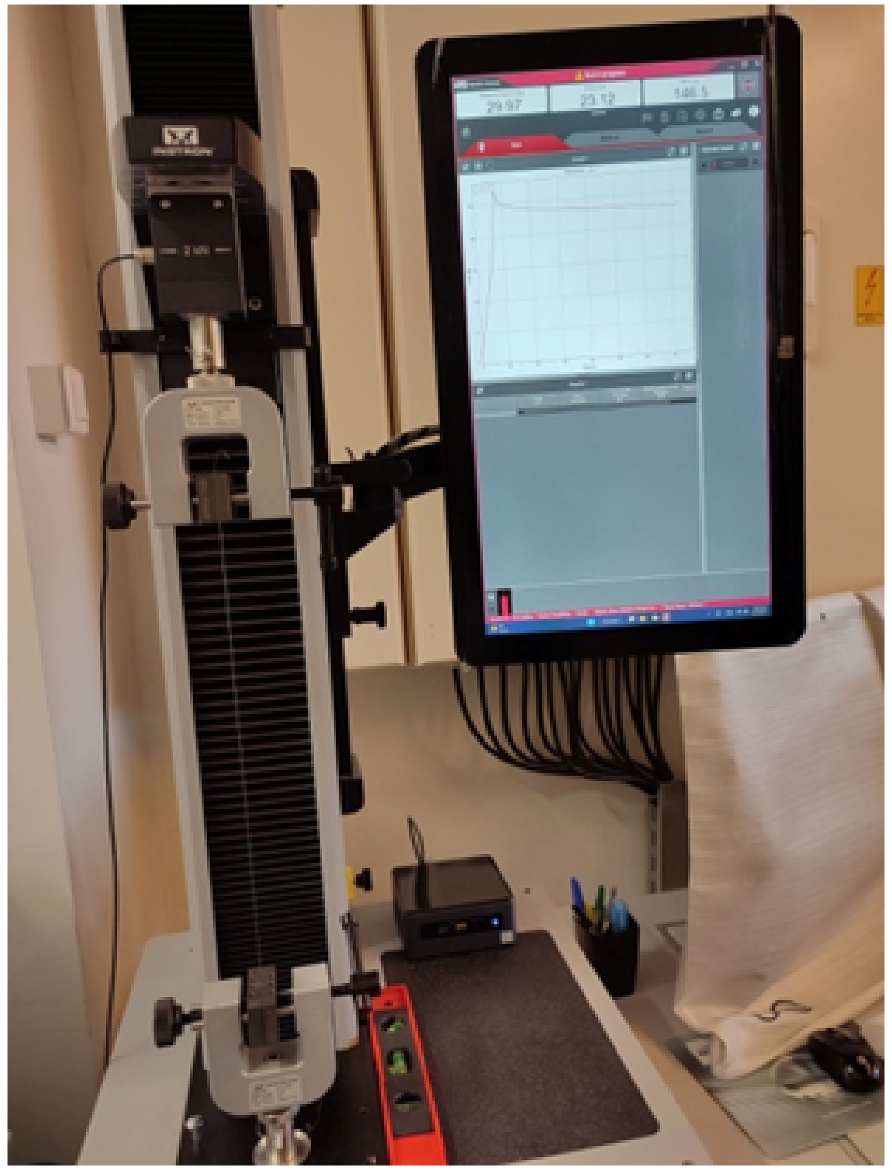


Fig. 1. Experimental apparatus.

2.1. Stress-relaxation experiment

- All samples had an initial length of 30 cm. The experimental apparatus is shown in Figure 1. During the first phase, the ramp phase, the samples were loaded under constant strain rate until a predetermined strain value of 10% was achieved.
- Three sample groups were subjected to different deformation rates, specifically 60 mm/min, 120 mm/min, and 360 mm/min, so that the tension phase lasted 30 s, 15 s, and 5 s, respectively, for these three sample groups.
- After that, the samples were exposed to a relaxation phase in which the achieved deformation was held constant for a period of 180 s for all samples.

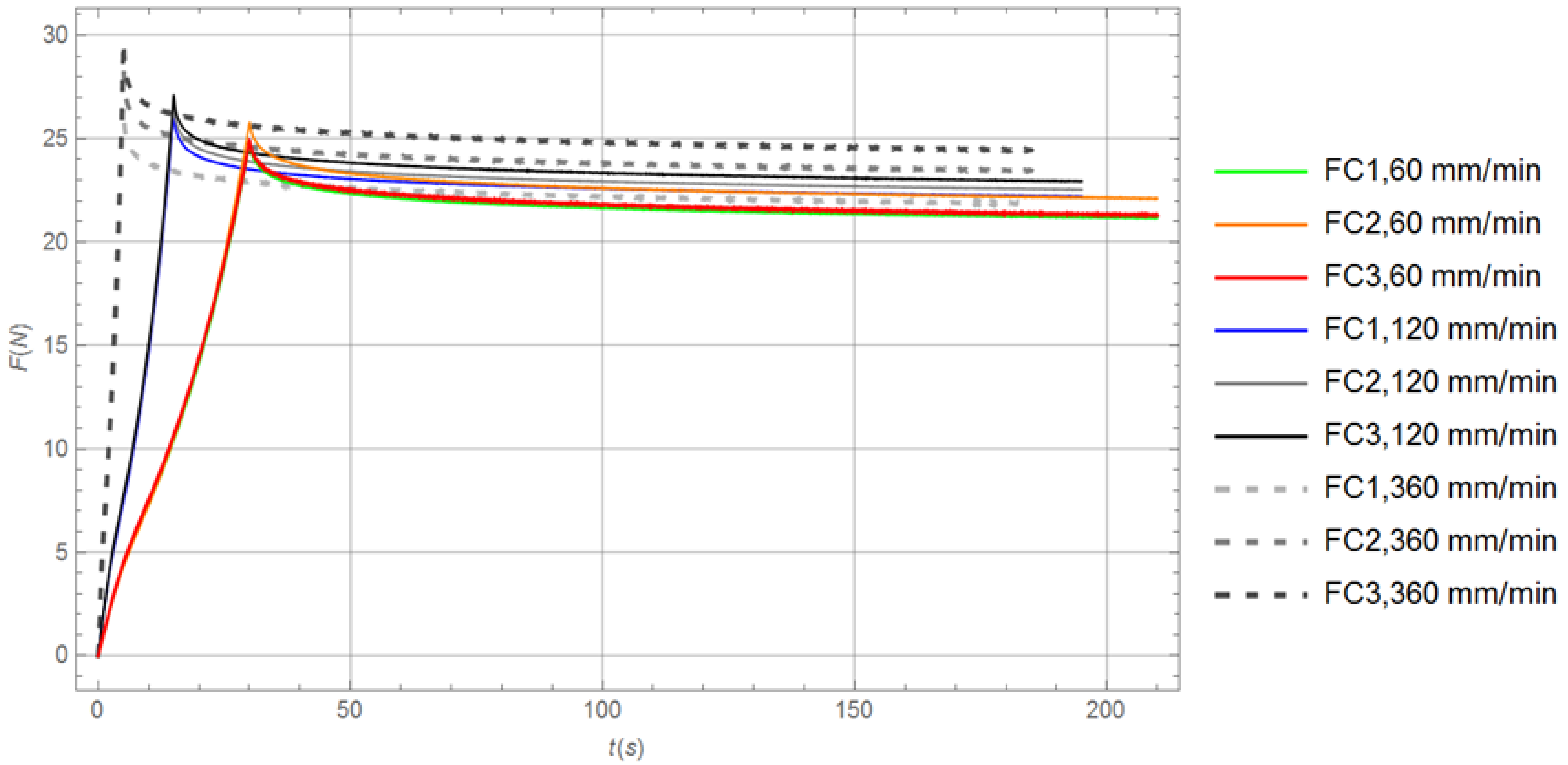


Fig. 2. Experimental results for nine fluorocarbon samples that were subjected to different deformation rates.

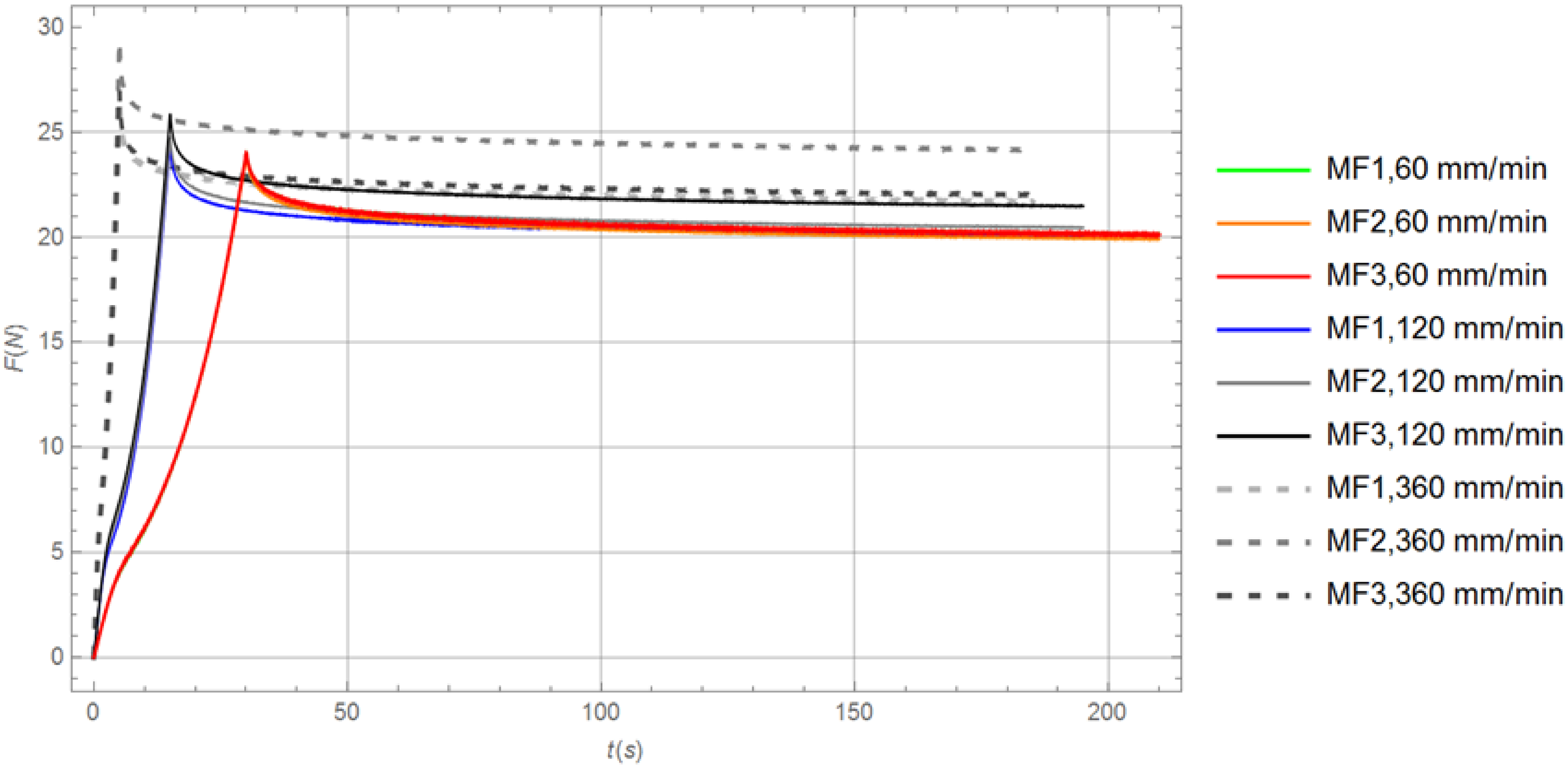


Fig. 3. Experimental results for nine monofilament samples that were subjected to different deformation rates.

2.Experiment and parameter identification

2.2. The model

The constitutive equation for Fractional Kelvin-Zener model: $f + \tau_{f\alpha} f^{(\alpha)} = E\left(x + \tau_{x\alpha} x^{(\alpha)}\right), \quad 0 < \alpha < 1,$

Standard Riemann–Liouville form: $[g(t)]^{(\alpha)} = \frac{\mathrm{d}^{\alpha}}{\mathrm{d}t^{\alpha}}[g(t)] = \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{1}{\Gamma(1-\alpha)}\int_0^t \frac{g(\tau)}{(t-\tau)^{\alpha}}\mathrm{d}\tau\right]$

Restrictions according to the second law of thermodynamics: $E > 0,\ \tau_{f\alpha} > 0,\ \tau_{x\alpha} > \tau_{f\alpha}$

2.Experiment and parameter identification

2.3. Numerical method

For a time step h: $z^{(\psi)} = h^{-\psi} \sum_{j=0}^{m} \omega_j^{(\psi)} z_{m-j},$ where $\omega_0^{(\psi)} = 1, \quad \omega_j^{(\psi)} = \left(1 - \frac{\psi + 1}{j}\right) \omega_{j-1}^{(\psi)}, \quad (j = 1, 2, 3, \ldots)$

From previous equations we obtain: $f_m + \tau_{f\alpha} h^{-\alpha} \sum_{j=0}^{m} \omega_j^{(\alpha)} f_{m-j} = E\left(x_m + \tau_{x\alpha} h^{-\alpha} \sum_{j=0}^{m} \omega_j^{(\alpha)} x_{m-j}\right)$

2.Experiment and parameter identification

2.3. Numerical method

Algorithm for obtaining the numerical solution:

$$f_m = \frac{1}{1+\tau_{f\alpha}h^{-\alpha}}\left\{Ex_m\left(1+\tau_{x\alpha}h^{-\alpha}\right)+h^{-\alpha}\sum_{j=1}^{m}\left[\omega_j^{(\alpha)}\left(E\tau_{x\alpha}x_{m-j}-\tau_{f\alpha}f_{m-j}\right)\right]\right\}$$

Final step, finding the minimum of the following function:

$$\vartheta\left(t,\alpha,E,\tau_{x\alpha},\tau_{f\alpha}\right)=\sum_{i=0}^{4}\left[f\left(t_i,\alpha,E,\tau_{x\alpha},\tau_{f\alpha}\right)-f_{EXPi}\right]^2$$

3.Results

From the obtained results, it can be noticed that there are no significant deviations in the parameter values within each group of three samples for both tested materials, indicating result repeatability.

Sample	α	E[N/mm]	$\tau_{x\alpha}[s^{\alpha}]$	$\tau_{f\alpha}[s^{\alpha}]$
FC1 (60 mm/min)	0.527	0.681	1.127	$4.882 \cdot 10^{-4}$
FC2 (60 mm/min)	0.529	0.712	1.104	$4.957 \cdot 10^{-4}$
FC3 (60 mm/min)	0.515	0.684	1.088	$4.958 \cdot 10^{-4}$
FC1 (120 mm/min)	0.437	0.71	0.671	$4.783 \cdot 10^{-4}$
FC2 (120 mm/min)	0.436	0.721	0.679	0.011
FC3 (120 mm/min)	0.447	0.734	0.862	0.153
FC1 (360 mm/min)	0.434	0.712	0.437	$2.971 \cdot 10^{-3}$
FC2 (360 mm/min)	0.43	0.763	0.431	$4.912 \cdot 10^{-4}$
FC3 (360 mm/min)	0.426	0.795	0.433	$4.912 \cdot 10^{-4}$

Table 1. Model parameter values for fluorocarbon

Sample	α	E[N/mm]	$\tau_{x\alpha}[s^{\alpha}]$	$\tau_{f\alpha}[s^{\alpha}]$
MF1 (60 mm/min)	0.585	0.652	1.463	$4.888 \cdot 10^{-4}$
MF2 (60 mm/min)	0.583	0.643	1.479	10^{-3}
MF3 (60 mm/min)	0.598	0.652	1.533	$4.962 \cdot 10^{-4}$
MF1 (120 mm/min)	0.513	0.648	0.898	$4.949 \cdot 10^{-4}$
MF2 (120 mm/min)	0.517	0.661	0.893	$4.95 \cdot 10^{-4}$
MF3 (120 mm/min)	0.514	0.694	0.86	$1.798 \cdot 10^{-5}$
MF1 (360 mm/min)	0.525	0.714	0.505	$2.466 \cdot 10^{-4}$
MF2 (360 mm/min)	0.532	0.797	0.481	$4.974 \cdot 10^{-4}$
MF3 (360 mm/min)	0.533	0.725	0.507	$4.971 \cdot 10^{-4}$

Table 2. Model parameter values for monofilament

4.Conclusions

Very similar behavior was observed within each sample group, as evidenced by the small deviations in the parameter values obtained for each group. The change in deformation speed affected the parameter values in a very similar manner for both materials. The highest and lowest deformation speeds for the tested samples were in a 6:1 ratio.

It would be interesting to investigate the effect of a much larger ratio between the maximum and minimum deformation speeds, and to analyze the potential application of relatively simple relaxation and creep experiments to collision problems where deformation speeds are very high.

Thank you for your attention!